Due: Tuesday March 8th, during class starting at noon

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1. A stick of proper length L moves past you at speed v. There is a time interval between the front end coinciding with you (first event) and the back end coinciding with you (second event). We are interested in this time interval and in the length of the stick, as seen from various frames.

   (a) **[2 pts.]** Draw snapshots of the two events as seen from your frame, marking with an arrow what is moving (you or the stick).

   (b) **[4 pts.]** Find the length of the stick as seen from your frame, and the time between the two events, as seen from your frame.

   (c) **[2 pts.]** Draw snapshots of the two events as seen from the frame of the stick, marking with an arrow what is moving, as seen from this frame.

   (d) **[4 pts.]** Find the length of the stick and the time between the two events, as seen from the frame of the stick.

   (e) **[2 pts.]** In which frame is the time interval the *proper* time interval: your frame or the stick's frame?

2. Frame S' moves at speed v_1 with respect to frame S, and frame S'' moves at speed v_2 with respect to frame S', both in the common x (or x' or x'') direction. The Lorentz transformations are

$$x' = \gamma_1(x - v_1 t)\ , \quad t' = \gamma_1(t - v_1 x/c^2) \qquad \text{with} \quad \gamma_1 = \gamma(v_1) = \frac{1}{\sqrt{1 - v_1^2/c^2}}$$

$$x'' = \gamma_2(x' - v_2 t')\ , \quad t'' = \gamma_2(t' - v_2 x'/c^2) \qquad \text{with} \quad \gamma_2 = \gamma(v_2) = \frac{1}{\sqrt{1 - v_2^2/c^2}}$$

   (a) **[4 pts.]** Express x'' (the coordinate of the S'' frame) in terms of x and t, in the form $x'' = Ax + Bt$. Express A and B in terms of v_1, v_2, γ_1 and γ_2.

   (b) **[2 pts.]** If frame S'' moves at speed u with respect to frame S, you would have $x'' = \gamma(u)(x - ut)$. Comparing with $x'' = Ax + Bt$, this implies $u = -B/A$. Using your expressions for A and B, find u. Have you recovered the relativistic velocity addition formula?

3. The following problems parctice using the velocity addition formula.

(a) **[4 pts.]** Two rockets approach each other, as observed from earth, each with speed v. What is the relative speed of one rocket as seen from the other?

(b) **[2 pts.]** Sketch the situation considered above in problem 3a in two ways: first from the earth's frame and then from the frame of one of the rockets. In both figures, mark each object (earth, first rocket, second rocket) with an arrow and expression, representing the direction and magnitude of the velocity of that object.

(c) **[4 pts.]** A car moves at speed v toward the left, while a light pulse moves with speed c toward the right. Use the velocity addition formula to find out, from the point of view of the car driver, how fast the light pulse approaches the car.

(d) **[4 pts.]** Spacecrafts A and B are traveling in the *same* direction with speeds $4c/5$ and $3c/5$ respectively, as seen from earth. (A is chasing B from behind.) Find the speed of A as seen from B.

(e) **[6 pts.]** Spacecraft C is between A and B, and traveling in the same direction. The passengers on C see the other two spacecrafts approaching C from opposite directions at the same speed. What is the speed of C as seen from earth? [Hint: You will need to solve a quadratic equation. This equation will have two solutions, of which only one is physically acceptable; mention why.]